

# Evaluación Bases de Datos: Diseño de Esquema

## Trabajo Práctico Intermedio

Curso: Backend Developer

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### Objetivo General

Es un sistema interno de gestión para una tienda de instrumentos musicales, utilizado por empleados, que permite cargar el catálogo de productos (marca, tipo, categoría) y registrar las ventas realizadas por el vendedor.

### II) Diagrama del modelo



### III) Justificación de embeber vs referenciar

#### Referencial

- Elegí la entidad Referenciada, porque puedo aprovechar mejor la gestión para una tienda de instrumentos musicales, su principal ventaja es una relación de datos de (“Uno a muchos”), esto evita principalmente la duplicidad de datos.
- Las colecciones son:
  - **INSTRUMENTS → BRANDS:** Porque necesito que traiga los nombres de los fabricantes de los instrumentos.
  - **INSTRUMENTS → TYPES:** Se necesita saber el tipo de instrumento es: Guitarra eléctrica, Bajo eléctrico, teclado, batería u otros tipos.
  - **TYPES → CATEGORIES:** Se necesita porque puedo saber, si el instrumento es de cuerdas, percusión o de viento. esto me va permitir de poder tener mayor modulabilidad, ya sea para reportes o indicadores y aprovechar mejor el frontend. con mayores funcionalidades.
  - **USERS, ROLES → AREAS:** Se puede separar a los empleados de acuerdo al rol que tiene y a que área que pertenece, un empleado administrativo puede hacer la carga de los instrumentos y un empleado vendedor puede atender a los clientes.
  - **INVOICES → USERS:** identifica qué empleado hizo la venta.
  - **INSTRUMENTS → USERS:** Permite identificar qué empleado cargó cada instrumento al catálogo.
  - **INVOICES\_ITEMS → INVOICES:** En una factura se puede vender muchos instrumentos, al separar el encabezado

“INVOICES” de los detalles “INVOICES\_ITEMS” esto evita que se duplique los datos y la información sea más consistente.

- **INVOICES\_ITEMS → INSTRUMENTS:** Identifica qué instrumentos fueron vendidos a partir que se cargó en **INVOICES\_ITEMS**.

#### IV) Esquema de las tablas

| TABLE    | FIELD        | TIPO DATO | TABLE REFERENCE | REQUIRED | VALIDACIONES                       | COMENTARIO                            |
|----------|--------------|-----------|-----------------|----------|------------------------------------|---------------------------------------|
| INVOICES | _id (PK)     | Objectid  |                 |          |                                    | n/a                                   |
|          | invoice_date | DATE      |                 | x        | default: Date.now                  | Graba fecha en momento de la creación |
|          | user         | Objectid  | ref: USERS._id  | x        |                                    | n/a                                   |
|          | client       | STRING    |                 | x        | minlength/maxlength/lowercase/trim | n/a                                   |
|          | total_amount | NUMBER    |                 | x        | min / max                          | n/a                                   |

| TABLE          | FIELD     | TIPO DATO | TABLE REFERENCE      | REQUIRED | VALIDACIONES | COMENTARIO |
|----------------|-----------|-----------|----------------------|----------|--------------|------------|
| INVOICES_ITEMS | _id (PK)  | Objectid  |                      |          |              |            |
|                | invoice   | Objectid  | ref: INVOICES._id    | x        |              | n/a        |
|                | product   | Objectid  | ref: INSTRUMENTS._id | x        |              | n/a        |
|                | amount    | NUMBER    |                      | x        | min / max    | n/a        |
|                | taxes     | NUMBER    |                      | x        | min / max    | n/a        |
|                | sub_total | NUMBER    |                      | x        | min / max    | n/a        |

| TABLE  | FIELD    | TIPO DATO | TABLE REFERENCE | REQUIRED | VALIDACIONES                             | COMENTARIO |
|--------|----------|-----------|-----------------|----------|------------------------------------------|------------|
| BRANDS | _id (PK) | Objectid  |                 |          |                                          | n/a        |
|        | name     | STRING    |                 | x        | minlength / maxlength/ unique /lowercase | n/a        |

| TABLE | FIELD    | TIPO DATO | TABLE REFERENCE     | REQUIRED | VALIDACIONES                            | COMENTARIO |
|-------|----------|-----------|---------------------|----------|-----------------------------------------|------------|
| TYPES | _id (PK) | Objectid  |                     |          |                                         | n/a        |
|       | name     | STRING    |                     | x        | minlength / maxlength/unique /lowercase | n/a        |
|       | category | Objectid  | ref: CATEGORIES._id |          |                                         | n/a        |

| TABLE      | FIELD    | TIPO DATO | TABLE REFERENCE | REQUIRED | VALIDACIONES                            | COMENTARIO |
|------------|----------|-----------|-----------------|----------|-----------------------------------------|------------|
| CATEGORIES | _id (PK) | Objectid  |                 |          |                                         | n/a        |
|            | name     | STRING    |                 | x        | minlength / maxlength/unique /lowercase | n/a        |

| TABLE | FIELD         | TIPO DATO | TABLE REFERENCE | REQUIRED | VALIDACIONES                            | COMENTARIO |
|-------|---------------|-----------|-----------------|----------|-----------------------------------------|------------|
| USERS | _id (PK)      | Objectid  |                 |          |                                         | n/a        |
|       | name          | STRING    |                 | x        | minlength / maxlength /lowercase        | n/a        |
|       | surname       | STRING    |                 | x        | minlength / maxlength /lowercase        | n/a        |
|       | rol           | Objectid  | ref: ROLES._id  | x        |                                         | n/a        |
|       | email         | STRING    |                 | x        | minlength / maxlength/unique /lowercase | n/a        |
|       | password_hash | STRING    |                 | x        | minlength / maxlength                   | n/a        |

| TABLE | FIELD    | TIPO DATO | TABLE REFERENCE | REQUIRED | VALIDACIONES                             | COMENTARIO |
|-------|----------|-----------|-----------------|----------|------------------------------------------|------------|
| ROLES | _id (PK) | Objectid  |                 |          |                                          | n/a        |
|       | name     | STRING    |                 | x        | minlength / maxlength/ unique /lowercase | n/a        |
|       | area     | Objectid  | ref: AREAS._id  | x        |                                          | n/a        |

| TABLE | FIELD    | TIPO DATO | TABLE REFERENCE | REQUIRED | VALIDACIONES                            | COMENTARIO |
|-------|----------|-----------|-----------------|----------|-----------------------------------------|------------|
| AREAS | _id (PK) | Objectid  |                 |          |                                         | n/a        |
|       | name     | STRING    |                 | x        | minlength / maxlength/unique /lowercase | n/a        |

| TABLE       | FIELD         | TIPO DATO | TABLE REFERENCE | REQUIRED | VALIDACIONES                                           | COMENTARIO                                                                                                                                     |
|-------------|---------------|-----------|-----------------|----------|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| INSTRUMENTS | _id (PK)      | ObjectId  |                 |          |                                                        | n/a                                                                                                                                            |
|             | brand         | ObjectId  | ref: BRANDS._id | x        |                                                        | n/a                                                                                                                                            |
|             | model         | STRING    |                 | x        | minlength /maxlength /lowercase /trim /lowercase       | n/a                                                                                                                                            |
|             | currency      | STRING    |                 | x        | enum: ["ARS", "USD"], default: "ARS" /uppercase: true  | Moneda del precio cargado                                                                                                                      |
|             | price         | NUMBER    |                 | x        | min / max                                              | n/a                                                                                                                                            |
|             | previousPrice | NUMBER    |                 |          | min                                                    | n/a                                                                                                                                            |
|             | type          | ObjectId  | ref: TYPES._id  | x        |                                                        | n/a                                                                                                                                            |
|             | available     | BOOLEAN   |                 | x        |                                                        | n/a                                                                                                                                            |
|             | condition     | STRING    |                 | x        | enum: ["new", "used"], default: "new" /lowercase: true | n/a                                                                                                                                            |
|             | color         | STRING    |                 | x        | minlength / maxlength /lowercase                       | n/a                                                                                                                                            |
|             | made          | STRING    |                 | x        | minlength / maxlength /lowercase                       | n/a                                                                                                                                            |
|             | sku           | STRING    |                 | x        | minlength / maxlength / unique /lowercase              | n/a                                                                                                                                            |
|             | description   | STRING    |                 | x        | minlength / maxlength /lowercase                       | n/a                                                                                                                                            |
|             | user          | ObjectId  | ref: USERS._id  | x        |                                                        | n/a                                                                                                                                            |
|             | create_date   | DATE      |                 |          | default: Date.now                                      | Graba fecha en creación                                                                                                                        |
|             |               |           |                 |          |                                                        | "El default solo aplica al crear el documento; para reflejar cada modificación posterior, la aplicación debe reasignar este valor manualmente" |
|             | update_date   | DATE      |                 |          | default: fecha actual                                  |                                                                                                                                                |
|             | is_delete     | BOOLEAN   |                 |          | default: false                                         | Baja logica                                                                                                                                    |

## V) Ejemplo json

### Table: AREAS

```
{
  "_id": "6aac8eec0d5926ec1690cfef",
  "name": "ventas"
}
{
  "_id": "51f3a7c92b8e4d106c9a2e17",
  "name": "deposito"
}
{
  "_id": "8c2d914fa63b7e0521d8f4a9",
  "name": "administracion"
}
```

### Table: BRANDS

```
{
  "_id": "5f8a3b1c9e6d4207a8b1c3d4",
  "name": "fender"
}
```

### Table: CATEGORIES

```
{
  "_id": "6aa2f4e9a0de7b789d1e37e2",
  "name": "cuerdas"
}
```

### Table: ROLES

```
{
  "_id": "6aa9682fa14db80c1ccc99aa",
  "name": "administración",
  "area": "8c2d914fa63b7e0521d8f4a9"
```

```

}
{
  "_id": "7bb1c2d3e4f5061728394a5b",
  "name": "vendedor",
  "area": "6aac8eec0d5926ec1690cfeb"
}
{
  "_id": "8cc2d3e4f5061728394a5b6c",
  "name": "deposito",
  "area": "51f3a7c92b8e4d106c9a2e17"
}

```

**Table: USERS**

```

{
  "_id": "65f1a2b3c4d5e6f7a8b9c0d1",
  "name": "juan",
  "surname": "pérez",
  "rol": "6aa9682fa14db80c1ccc99aa",
  "email": "juan.perez@example.com",
  "password_hash": "$2b$10$EixZaYVK1fsbw1ZfbX3OXePaWxn96p36WQOEg6Lruj"
}
{
  "_id": "76f2b3c4d5e6f708192a3b4c",
  "name": "morena",
  "surname": "hererro",
  "rol": "7bb1c2d3e4f5061728394a5b",
  "email": "morena.herro@example.com",
  "password_hash": "$2b$10$KpX1mZbQvN8Rj3sT9yLoWeHtG6cF0aB2dE4xC7uJ5nM1pQrS"
}
{
  "_id": "87f3c4d5e6f708192a3b4c5d",
  "name": "carlos",
  "surname": "novac",
  "rol": "8cc2d3e4f5061728394a5b6c",
  "email": "carlos.novac@example.com",
  "password_hash": "$2b$10$LqY2nAcRwO9Sk4tU0zMpXfluH7dG1bC3eF5yD8vK6oN2qRsT"
}

```

**Table: INSTRUMENTS**

```

{
  "_id": "65f1b8a9c4d5e6f7a8b9c101",
  "brand": "5f8a3b1c9e6d4207a8b1c3d4",
  "model": "stratocaster standard",
  "currency": "ARS",
  "price": 1200,
  "previousPrice": 1350,
  "type": "65f1b8a9c4d5e6f7a8b9c103",
  "available": true,
  "condition": "new",
  "color": "sunburst",
  "made": "usa",

```

```
"sku": "fender-strat-001",
"description": "guitarra eléctrica con tres pastillas single-coil",
"user": "65f1a2b3c4d5e6f7a8b9c0d1",
"create_date": "2026-03-10T10:00:00.000Z",
"update_date": "2026-03-15T12:30:00.000Z",
"is_delete": false
}
```

**Table: INVOICES**

```
{
  "_id": "65f2c910c4d5e6f7a8b9c201",
  "invoice_date": "2026-03-17T15:30:00.000Z",
  "user": "65f1a2b3c4d5e6f7a8b9c0d1",
  "client": "juan perez",
  "total_amount": 1452.70
}
```

**Table: INVOICES\_ITEMS**

```
{
  "_id": "65f3d021c4d5e6f7a8b9c301",
  "invoice": "65f2c910c4d5e6f7a8b9c201",
  "product": "65f1b8a9c4d5e6f7a8b9c101",
  "amount": 1,
  "taxes": 252.70,
  "sub_total": 1200
}
```

**Table: TYPES**

```
{
  "_id": "65f1b8a9c4d5e6f7a8b9c103",
  "name": "guitarra eléctrica",
  "category": "6aa2f4e9a0de7b789d1e37e2"
}
```

**Table: BRANDS**

```
{
  "_id": "3d9e1f2a4b5c6d7e8f9a0b1c",
  "name": "gibson"
}
{
  "_id": "4e0f2a3b5c6d7e8f9a0b1c2d",
  "name": "yamaha"
}
```

**Table: CATEGORIES**

```
{
  "_id": "5f1a3b4c6d7e8f9a0b1c2d3e",
  "name": "viento"
}
{
  "_id": "6a2b4c5d7e8f9a0b1c2d3e4f",
  "name": "percusion"
}
```

```
}
```

**Table: TYPES**

```
{
```

```
  "_id": "7b3c5d6e8f9a0b1c2d3e4f5a",
  "name": "saxofon",
  "category": "5f1a3b4c6d7e8f9a0b1c2d3e"
```

```
}
```

```
{
```

```
  "_id": "8c4d6e7f9a0b1c2d3e4f5a6b",
  "name": "bateria",
  "category": "6a2b4c5d7e8f9a0b1c2d3e4f"
```

```
}
```

**Table: INSTRUMENTS**

```
{
```

```
  "_id": "9d5e7f8a0b1c2d3e4f5a6b7c",
  "brand": "3d9e1f2a4b5c6d7e8f9a0b1c",
  "model": "les paul standard",
  "currency": "ARS",
  "price": 1800,
  "previousPrice": 1900,
  "type": "65f1b8a9c4d5e6f7a8b9c103",
  "available": true,
  "condition": "new",
  "color": "negro",
  "made": "usa",
  "sku": "gibson-lespaul-002",
  "description": "guitarra eléctrica de cuerpo sólido con dos humbuckers",
  "user": "65f1a2b3c4d5e6f7a8b9c0d1",
  "create_date": "2026-04-02T09:15:00.000Z",
  "update_date": "2026-04-02T09:15:00.000Z",
  "is_delete": false
```

```
}
```

```
{
```

```
  "_id": "0e6f8a9b1c2d3e4f5a6b7c8d",
  "brand": "4e0f2a3b5c6d7e8f9a0b1c2d",
  "model": "yts-280",
  "currency": "ARS",
  "price": 950,
  "type": "7b3c5d6e8f9a0b1c2d3e4f5a",
  "available": true,
  "condition": "new",
  "color": "dorado",
  "made": "japon",
  "sku": "yamaha-sax-003",
  "description": "saxofon alto estudiantil",
  "user": "65f1a2b3c4d5e6f7a8b9c0d1",
  "create_date": "2026-04-05T11:00:00.000Z",
  "update_date": "2026-04-05T11:00:00.000Z",
  "is_delete": false
```

```
}
```

**Table: INVOICES**

```
{
  "_id": "1e6f8a9b0c1d2e3f4a5b6c7d",
  "invoice_date": "2026-04-03T14:00:00.000Z",
  "user": "76f2b3c4d5e6f708192a3b4c",
  "client": "aida felicinda",
  "total_amount": 2124
}
{
  "_id": "2f7a9b0c1d2e3f4a5b6c7d8e",
  "invoice_date": "2026-04-06T16:45:00.000Z",
  "user": "87f3c4d5e6f708192a3b4c5d",
  "client": "nuria lord",
  "total_amount": 1092.50
}
```

**Table: INVOICES\_ITEMS**

```
{
  "_id": "3a8b0c1d2e3f4a5b6c7d8e9f",
  "invoice": "1e6f8a9b0c1d2e3f4a5b6c7d",
  "product": "9d5e7f8a0b1c2d3e4f5a6b7c",
  "amount": 1,
  "taxes": 324,
  "sub_total": 1800
}
{
  "_id": "4b9c1d2e3f4a5b6c7d8e9f0a",
  "invoice": "2f7a9b0c1d2e3f4a5b6c7d8e",
  "product": "0e6f8a9b1c2d3e4f5a6b7c8d",
  "amount": 1,
  "taxes": 142.50,
  "sub_total": 950
}
```